

MSCA STAFF EXCHANGES

Marie Skłodowska-Curie Staff Exchanges

Cross-Sector Exchange for Precision Fermentation in Sustainable Food AGRISWAP

Field	Value
Call Identifier	HORIZON-MSCA-2026-SE-01-01
Type of Action	MSCA Staff Exchanges
Duration	48 months
TRL Range	TRL 3 to TRL 6
Total Budget	EUR 1,080,000
EU Contribution	EUR 1,080,000

Abstract

AGRISWAP (Cross-Sector Exchange for Precision Fermentation in Sustainable Food) is a 48-month MSCA Staff Exchanges project that builds a durable, inter-sectoral, and international research and innovation network to accelerate the knowledge transfer and industrial readiness of precision fermentation for sustainable food production. The consortium unites Wageningen University & Research (Netherlands, coordinator), Instituto Superior de Agronomia – Universidade de Lisboa (Portugal), FermentTech Italia Srl (Italy, SME), and EMBRAPA Food Technology (Brazil, third country). Through up to 360 funded person-months of physical secondments—targeting the completion of at least 300 by month 42—structured around four thematic clusters (strain engineering and synthetic biology, fermentation process optimisation and scale-up, ingredient characterisation and regulatory science, and knowledge exploitation), AGRISWAP advances microbial cell-factory know-how from TRL 3 to TRL 6 by embedding frontier academic expertise within industrial production environments and vice versa. At least 20 staff members at all career stages benefit from immersive cross-sectoral and cross-cultural secondment experiences, enhancing their skills, inter-sectoral employability, and entrepreneurial mind-sets. Open science commitments are operationalised through a minimum of 12 open-access publications, FAIR-compliant dataset deposition, and a Data Management Plan maintained throughout the project. Societal and policy impact is delivered through two open stakeholder workshops, a regulatory roadmap and policy brief for EU novel-food authorities, and a signed post-project Joint Research Agreement committing partners to sustained collaboration. A gender-action plan ensures at least 40% representation of the under-represented gender among seconded staff. AGRISWAP directly addresses EU Farm to Fork Strategy priorities and Horizon Europe bioeconomy goals, generating the scientific, human, and institutional capital needed to position Europe as a global leader in sustainable food biotechnology.

Keywords: precision fermentation, sustainable food systems, inter-sectoral mobility, staff secondments, bioprocess engineering, synthetic biology, food safety, knowledge transfer, MSCA, international collaboration

SECTION 1 - EXCELLENCE

1.1 Objectives and Ambition

AGRISWAP establishes a durable, inter-sectoral, and international research and innovation network centred on precision fermentation as a lever for sustainable food systems. The consortium unites Wageningen University & Research (WUR, Netherlands) as coordinator, a Portuguese academic institute (Instituto Superior de Agronomia, Universidade de Lisboa), an Italian food-technology SME (FermenTech Italia Srl), and a Brazilian non-academic research centre (EMBRAPA Food Technology, Brazil). This four-partner configuration fulfils and exceeds the MSCA Staff Exchanges eligibility requirements: it spans three EU/Associated Countries, integrates academic and non-academic sectors, and includes a non-associated third-country organisation, thereby maximising the international dimension explicitly encouraged by the Work Programme.

The overarching scientific objective of AGRISWAP is to advance the knowledge base and industrial readiness of microbial cell-factory design, fermentation process optimisation, and downstream bioprocessing for the production of functional food ingredients. Four specific, measurable, and time-bound objectives structure the programme: (O1) transfer and co-develop strain-engineering methodologies between academic and industrial partners; (O2) validate scalable precision-fermentation protocols for at least two target functional ingredients by month 42; (O3) build cross-sectoral competence in regulatory science and food-safety assessment for novel fermentation-derived products; and (O4) establish a lasting collaborative governance structure that sustains joint R&I beyond the project lifetime through a signed Joint Research Agreement. These objectives are ambitious yet tractable within 48 months, and they are firmly anchored in EU priorities for sustainable food systems, the Farm to Fork Strategy, and the Horizon Europe Mission on Soil Health and Food.

The project advances the state of the art in two distinct ways. Scientifically, AGRISWAP moves precision fermentation from proof-of-concept (TRL 3) to validated pilot-scale demonstration (TRL 6) by integrating synthetic-biology strain design with industrial bioprocess engineering—a combination currently fragmented across separate research silos. Structurally, the exchange programme embeds knowledge through hands-on practice at partner facilities, generating tacit expertise that publications and workshops alone cannot transfer. The scientific quality of AGRISWAP rests on the genuine complementarity of partner expertise: WUR contributes frontier knowledge in systems biology and metabolic engineering; the Portuguese partner brings food-science and fermentation-process competence; FermenTech Italia provides industrial-scale bioprocess engineering and product-validation infrastructure; and EMBRAPA contributes access to unique tropical microbial biodiversity and long-standing expertise in agri-food biotechnology. No single partner possesses this full knowledge stack, making the exchange of staff indispensable rather than incidental to achieving the objectives.

1.2 Relation to the Work Programme

AGRISWAP responds directly to the HORIZON-MSCA-2026-SE-01-01 call, which targets innovative international, inter-sectoral, and interdisciplinary collaboration through staff secondments that generate or strengthen long-term partnerships. The project addresses all three mobility dimensions identified in the Work Programme: international (EU–Brazil exchanges), inter-sectoral (academia–SME–non-academic research centre), and interdisciplinary (microbiology, food science, bioprocess engineering, regulatory affairs). The emphasis on knowledge transfer through physical secondments, structured mentoring, and joint training activities aligns precisely with the expected outcomes for both individual staff members and participating organisations as set out in the call text.

AGRISWAP also contributes to the broader Horizon Europe goals of strengthening the European

Research and Innovation Area, enhancing Europe's attractiveness as an R&I destination, and fostering an open-science and entrepreneurial culture. The participation of FermenTech Italia as an SME reflects the Work Programme's explicit inclusion of non-academic actors, and the engagement of EMBRAPA (Brazil) exemplifies the encouraged participation of non-associated third countries, introducing a genuine South–North knowledge dimension. The precision-fermentation thematic focus aligns with European strategic priorities in sustainable bioeconomy and food-system resilience, ensuring policy coherence across Horizon Europe pillars and with the Farm to Fork Strategy.

1.3 Concept and Methodology

The methodological foundation of AGRISWAP is a structured secondment programme organised around four thematic research clusters: (i) strain engineering and synthetic biology, (ii) fermentation process optimisation and scale-up, (iii) ingredient characterisation, safety, and regulatory compliance, and (iv) techno-economic and life-cycle analysis. Staff from each partner are seconded to one or more counterpart organisations to perform hands-on research tasks within these clusters, ensuring that knowledge is embedded through practice rather than transferred solely through documents or workshops. This approach is explicitly designed to go beyond the state of the art: by coupling cutting-edge synthetic-biology workflows (WUR) with industrial design-of-experiments at pilot scale (FermenTech Italia) and with novel tropical substrate screening (EMBRAPA), the programme creates methodological combinations that no individual partner could develop in isolation.

The secondment programme is complemented by joint training events (annual consortium schools), a shared digital laboratory notebook for open knowledge management, and regular virtual seminars between physical secondments. A gender-inclusive recruitment approach is embedded throughout the secondment planning: the consortium commits to gender balance across seconded staff cohorts, consistent monitoring via a gender-action plan activated from month 1, and targeted mentoring for early-career researchers regardless of gender. Open science principles are operationalised through a Data Management Plan submitted at month 6, mandatory deposition of experimental datasets in FAIR-compliant repositories, and an open-access publication mandate for all peer-reviewed outputs. Intellectual property arising from joint activities is governed by a Consortium Agreement concluded before month 3, distinguishing background IP from foreground IP and providing transparent licensing rules that protect both academic and industrial partners.

The methodology is designed to be adaptive: a mid-term review at month 24 assesses secondment flows, research progress, and emerging needs, allowing reallocation of person-months within the agreed maximum of 360 across the consortium. This built-in flexibility, combined with robust project management (WP5), ensures that the programme remains on track even if individual secondments are delayed by unforeseen circumstances such as visa processing times or temporary travel restrictions.

1.4 Interdisciplinary Considerations

Precision fermentation is intrinsically interdisciplinary, requiring the integration of molecular biology, microbial genetics, chemical and bioprocess engineering, food science, nutrition, and regulatory affairs. AGRISWAP explicitly exploits this breadth as a scientific resource: researchers seconded from WUR (systems biology) to FermenTech Italia (bioprocess engineering) encounter and must master industrial scale-up logic, translating mechanistic biological knowledge into robust manufacturing parameters. Conversely, staff from the Portuguese partner (food science) seconded to EMBRAPA (tropical microbial diversity) integrate ecological microbiology into food-ingredient development pipelines, broadening the feedstock base available to European industry. This cross-disciplinary exposure is a deliberate design feature, not an incidental by-product.

The consortium further integrates social-science perspectives through the techno-economic and life-

cycle analysis cluster, which requires engagement with sustainability science and consumer-behaviour research to ensure that validated fermentation processes are genuinely viable at system level. Joint training events include dedicated modules on science communication and responsible research and innovation (RRI), equipping seconded staff to work effectively across disciplinary and sectoral boundaries and to engage diverse stakeholders. The interdisciplinary approach is expected to generate novel methodological combinations—most notably the coupling of synthetic-biology design tools with traditional fermentation process knowledge and with regulatory-science frameworks—that neither discipline could achieve independently, thereby producing genuine scientific added value beyond the sum of the partners' individual capabilities.

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SECTION 2 - IMPACT

2.1 Pathways Towards Impact

AGRISWAP generates impact at three interconnected levels, each with concrete, trackable outcomes.

For individual staff members, secondments provide immersive exposure to complementary research cultures, techniques, and career trajectories. Early-career researchers gain experience in both academic and industrial environments, directly enhancing their employability across sectors; the target is that at least 20 individuals at all career stages complete a cross-sectoral secondment by month 48. Senior researchers build international co-supervision networks and deepen their translation competences. Administrative and technical staff participating in secondments acquire project-management, quality-assurance, and regulatory-science skills that strengthen their home institutions upon return. All seconded staff benefit from structured mentoring and from sustained inclusion in a network that outlasts the funded period.

For participating organisations, AGRISWAP builds durable institutional capacity. WUR and the Portuguese partner deepen their industrial-translation pipelines, creating direct pathways from academic discovery to pilot-scale validation. FermenTech Italia gains access to frontier academic knowledge and to a Brazilian supply chain of novel microbial strains, reducing its strain-sourcing costs and diversifying its innovation portfolio. EMBRAPA strengthens its linkages to European food-technology markets and regulatory frameworks, increasing the commercial relevance of its biodiversity assets. The long-term collaboration is institutionalised through a Joint Research Agreement signed at month 42, committing all partners to at least two post-project collaborative funding applications and to maintaining a shared secondment pathway for at least three years beyond project end.

At the societal level, AGRISWAP contributes to food-system sustainability by advancing the industrial readiness of precision-fermentation technologies—from TRL 3 to TRL 6—that can reduce the environmental footprint of food production relative to conventional animal-derived or chemically synthesised ingredients. The validated protocols and open-access datasets generated during secondments will be freely available to the wider R&I community, multiplying impact well beyond the consortium and directly supporting the EU Farm to Fork Strategy targets for sustainable food systems.

2.2 Dissemination, Exploitation and Communication

The dissemination and exploitation strategy targets three distinct audiences with tailored, proportionate activities, coordinated by WUR's communication office and reviewed annually by the Supervisory Board.

For the scientific community, the consortium commits to at least 12 open-access peer-reviewed publications across the project lifetime, four conference presentations per year at major food-biotechnology and bioprocess-engineering venues, and the deposition of all experimental datasets in FAIR-compliant public repositories under accession numbers logged in the Data Management Plan. Where secondment activities generate protectable foreground intellectual property, a joint patent or utility-model application will be filed, with FermenTech Italia leading industrial exploitation and scale-up.

For industry and policy stakeholders, AGRISWAP will organise two open stakeholder workshops (months 18 and 36) bringing together representatives from European food industry associations, national food-safety agencies, and the EFSA advisory network, with a target of at least 60 external participants in total. A policy brief summarising the regulatory implications of precision-fermentation

ingredients under EU Regulation 2015/2283 will be produced at month 40 and actively distributed to relevant authorities and industry bodies.

For the general public and educational audiences, the consortium will maintain a multilingual project website (six languages, launched by month 3), produce a short video series on precision fermentation and sustainable food, and engage in at least four science-communication events per partner over the project lifetime. These activities are designed to build public understanding of and trust in precision-fermentation technologies, contributing to broader societal acceptance that is a prerequisite for the commercial impact targeted by the project.

2.3 Summary of Impact Indicators

Indicator	Target	Measurement
Peer-reviewed open-access publications	12 over 48 months	DOI records in OpenAIRE; annual count reported in progress reports
Person-months of secondment completed	300 out of 360 maximum by month 42	Secondment records in the grant management system; verified against timesheets
Staff members completing cross-sectoral secondment	At least 20 individuals across all career stages	Individual mobility logs; HR records at sending and hosting institutions
Gender balance in seconded staff	At least 40% representation of the under-represented gender	Annual gender-action plan monitoring report submitted to Project Officer
Joint post-project funding applications	At least 2 applications submitted within 24 months of project end	Copies of submitted applications provided to consortium coordinator
Stakeholder workshop participants	At least 60 external participants across two workshops	Registration records; post-event survey reports
Datasets deposited in FAIR repositories	At least 8 datasets by month 48	DOI/accession numbers listed in Data Management Plan updates

SECTION 3 - QUALITY AND EFFICIENCY OF THE IMPLEMENTATION

3.1 Work Plan and Work Packages

WP	Title	Lead	PM	Duration
WP1	Strain Engineering and Synthetic Biology Secondments	Wageningen University & Research	80	M1-M36
WP2	Fermentation Process Optimisation and Scale-Up	FermenTech Italia Srl	80	M6-M42
WP3	Ingredient Characterisation, Safety, and Regulatory Science	Instituto Superior de Agronomia, Universidade de Lisboa	60	M12-M48
WP4	Dissemination, Communication, and Knowledge Exploitation	Wageningen University & Research	30	M1-M48
WP5	Project Management, Ethics, and Sustainability	Wageningen University & Research	50	M1-M48

WP1: Strain Engineering and Synthetic Biology Secondments

Lead: Wageningen University & Research | Person-months: 80 | Duration: M1 - M36

WP1 coordinates secondments focused on microbial cell-factory design, CRISPR-based genome editing, and metabolic flux analysis—the foundational scientific activities underpinning AGRISWAP's TRL advancement. WUR researchers are seconded to FermenTech Italia for industrial strain validation at pilot scale, directly bridging the gap between academic strain design and industrial production requirements. EMBRAPA staff are hosted at WUR for advanced synthetic-biology training, enabling the screening of novel tropical microbial isolates with state-of-the-art tools. Knowledge generated during secondments is captured in jointly authored laboratory protocols deposited in the shared digital notebook, and in at least three co-authored peer-reviewed publications. WP1 feeds directly into WP2 by providing validated strains to the process-optimisation pipeline.

T1.1: Metabolic Engineering Secondments (WUR to FermenTech) - Two WUR researchers (total 12 person-months) are seconded to FermenTech Italia to transfer CRISPR-based metabolic-engineering workflows into an industrial context and validate engineered strains at pilot scale. Secondees work alongside FermenTech bioprocess engineers, ensuring that strain-design decisions are informed by industrial fermentation constraints from the outset. Progress is reviewed quarterly by the Scientific Coordination Committee.

T1.2: Tropical Biodiversity and Strain Prospecting (EMBRAPA to WUR) - EMBRAPA researchers (8 person-months) are seconded to WUR to apply synthetic-biology screening and high-throughput characterisation tools to a curated library of tropical microbial isolates, expanding the strain portfolio available to all consortium partners. Selected strains with favourable production phenotypes are incorporated into the WP2 fermentation-optimisation pipeline.

T1.3: Joint Protocol Development and IP Documentation - All T1.1 and T1.2 secondees collaboratively draft standardised, transferable strain-engineering protocols. The Consortium Agreement's IP provisions are applied to document background and foreground IP arising from joint strain development, ensuring clear ownership records before any results are disclosed externally.

WP2: Fermentation Process Optimisation and Scale-Up

Lead: FermenTech Italia Srl | Person-months: 80 | Duration: M6 - M42

WP2 governs secondments addressing bioprocess design-of-experiments, fed-batch and continuous fermentation optimisation, and scale-up from laboratory to pilot-to-demo scale. FermenTech Italia, as the sole industrial partner with dedicated pilot-scale infrastructure, leads this work package and hosts academic staff from WUR and the Portuguese partner. FermenTech engineers are in turn seconded to academic partners for scale-down modelling and predictive process design, creating a genuine bidirectional knowledge flow. The concrete deliverable is the validated pilot-scale production protocol for at least two target functional ingredients by month 42, representing the

project's primary TRL advancement milestone.

T2.1: Academic-to-Industry Process Optimisation Secondments - Researchers from WUR and the Portuguese partner are seconded to FermenTech Italia (20 person-months combined) to co-develop statistically optimised fed-batch fermentation protocols for two target functional ingredients. Secondees gain direct experience of industrial quality-assurance requirements, Good Manufacturing Practice environments, and regulatory documentation standards.

T2.2: Industry-to-Academia Scale-Down Modelling Secondments - FermenTech bioprocess engineers are seconded to WUR (12 person-months) to transfer industrial scale-up logic and empirical process knowledge into academic predictive-modelling and digital-twin frameworks. The resulting models accelerate future design cycles by reducing the number of costly pilot-scale experiments required.

T2.3: Brazil–Europe Process Benchmarking - EMBRAPA staff are seconded to FermenTech Italia (8 person-months) to benchmark fermentation processes using tropical substrates and novel EMBRAPA-sourced strains against European industrial standards. This activity identifies substrate-sourcing opportunities that could diversify European fermentation supply chains and reduce input costs.

WP3: Ingredient Characterisation, Safety, and Regulatory Science

Lead: Instituto Superior de Agronomia, Universidade de Lisboa | Person-months: 60 | Duration: M12 - M48

WP3 focuses on the analytical characterisation of fermentation-derived ingredients, food-safety testing, and regulatory-pathway mapping for the EU novel-food framework (Regulation 2015/2283). The Portuguese partner leads, leveraging its accredited food-safety testing infrastructure and established expertise in EU novel-food dossier preparation. FermenTech technical staff and EMBRAPA researchers are hosted for safety characterisation work; Portuguese partner staff are seconded to WUR for advanced omics-based safety assessment methodologies. The work package ensures that the technical outputs of WP1 and WP2 are accompanied by the regulatory intelligence required for market entry, directly connecting scientific validation to commercial impact.

T3.1: Analytical and Safety Characterisation Secondments - FermenTech technical staff are seconded to the Portuguese partner (10 person-months) to access accredited food-safety testing facilities and to build in-house regulatory-science capacity within the SME. Secondees conduct compositional analysis, allergenicity and toxicology assessments, and stability testing of pilot-scale ingredient batches produced in WP2.

T3.2: Regulatory Pathway Mapping and Policy Brief Drafting - A joint secondment activity involving Portuguese partner and WUR staff (8 person-months combined) produces a comprehensive regulatory roadmap for precision-fermentation ingredients under EU Regulation 2015/2283, including pre-submission consultation strategies and data-package requirements. The regulatory roadmap and an associated policy brief are the primary public-facing outputs of this task, targeting food-safety agencies, industry associations, and EFSA advisory bodies.

WP4: Dissemination, Communication, and Knowledge Exploitation

Lead: Wageningen University & Research | Person-months: 30 | Duration: M1 - M48

WP4 manages all outward-facing activities, ensuring that the scientific and technical outputs of WP1 –WP3 reach and benefit stakeholders beyond the consortium. Activities include the open-access publication pipeline, stakeholder workshops, the multilingual project website, public science-communication events, and the identification and exploitation of protectable results. Short administrative secondments support cross-partner coordination of dissemination and communication tasks, building a shared communication culture across the diverse consortium. WP4 is co-designed with WP5 to ensure that exploitation activities are consistent with the Consortium Agreement's IP provisions.

T4.1: Open-Access Publication and Dataset Management - WUR coordinates the manuscript submission pipeline across all work packages, ensuring compliance with the open-access mandate and timely deposition of datasets in FAIR-compliant repositories. Annual bibliometric reports track publication outputs against targets, and DOI/accession numbers are logged in successive Data Management Plan updates submitted under WP5.

T4.2: Stakeholder Workshops and Science Communication - Two open stakeholder workshops (months 18 and 36) are organised, each targeting at least 30 external participants from industry, regulatory bodies, and civil society. Annual public-engagement events are held at each partner institution. The multilingual project website (six languages) and a short video series on precision fermentation and sustainable food are produced and maintained throughout the project lifetime.

WP5: Project Management, Ethics, and Sustainability

Lead: Wageningen University & Research | Person-months: 50 | Duration: M1 - M48

WP5 provides the governance framework that underpins the entire project. It establishes a

Supervisory Board (strategic oversight, meeting bi-annually), a Scientific Coordination Committee (operational research coordination, meeting quarterly), and a dedicated Project Manager housed at WUR for day-to-day administration. WP5 covers ethics compliance, gender-action plan monitoring, Data Management Plan maintenance, financial reporting to the European Commission, and mid-term review. Crucially, WP5 also oversees the preparation and signature of the post-project Joint Research Agreement, which is the primary instrument for institutionalising the collaboration beyond the funded period. Effective WP5 management is a prerequisite for the timely initiation of secondments and the reliable delivery of all other work packages.

T5.1: Governance and Financial Management - The Project Manager at WUR provides day-to-day coordination, prepares financial and progress reports for the European Commission, maintains the secondment tracking system, and ensures that all partner reporting obligations are met on schedule. The Consortium Agreement and any amendments are managed under this task.

T5.2: Ethics, Gender, and Open Science Compliance - Annual ethics self-assessments are conducted in accordance with Horizon Europe requirements. The gender-action plan is activated from month 1, with annual progress reviews submitted to the Project Officer. Data Management Plan updates are submitted at months 6, 24, and 48. Any ethics issues arising from secondment activities—including data privacy, dual-use considerations, and research involving human subjects in food-safety testing—are addressed through the established ethics committee procedure.

T5.3: Mid-Term Review and Long-Term Sustainability Planning - An internal mid-term review at month 24 assesses secondment flows, scientific progress against milestones, financial expenditure, and the gender balance of seconded staff cohorts. Findings are used to update the secondment plan and reallocate person-months where necessary. The Joint Research Agreement is drafted from month 36 and finalised for signature by all partners at month 42, committing the consortium to sustained collaboration and at least two joint post-project funding applications.

3.2 Consortium Composition

Organisation	Country	Type	Role	Description
Wageningen University & Research	Netherlands	Academic	Coordinator	Consortium Coordinator; leads WP1, WP4, WP5; hosts incoming secondees in synthetic biology and process modelling
Instituto Superior de Agronomia, Universidade de Lisboa	Portugal	Academic	Partner	Leads WP3; provides food-science and food-safety expertise; hosts industrial and third-country secondees
FermenTech Italia Srl	Italy	Non-Academic (SME)	Partner	Leads WP2; provides industrial bioprocess infrastructure; hosts academic secondees; leads IP exploitation
EMBRAPA Food Technology	Brazil	Non-Academic (Research Centre, Third Country)	Partner	Contributes tropical microbial biodiversity and agri-food bioprocess expertise; participates in WP1 and WP2 secondments

The AGRISWAP consortium is uniquely constituted to deliver the project objectives, and its composition represents genuine added value that no smaller or more homogeneous grouping could replicate. Each partner occupies a non-overlapping, world-class niche: WUR's depth in systems biology and metabolic engineering, the Portuguese partner's accredited food-safety infrastructure and novel-food regulatory expertise, FermenTech Italia's industrial pilot-scale bioprocess capability and product-validation experience, and EMBRAPA's unparalleled access to tropical microbial biodiversity and its longstanding agri-food biotechnology networks. These capabilities are

individually strong and collectively irreplaceable for the TRL 3-to-6 advancement targeted by the project.

The inter-sectoral dimension—two academic institutions, one SME, and one non-academic public research centre—is a structural feature of the consortium, not a cosmetic addition. It ensures that secondments generate bidirectional, sector-crossing knowledge transfer rather than simple information exchange between similar organisations. The international dimension—three EU/Associated Countries plus Brazil—introduces a South–North axis of scientific exchange that enriches the European R&I base with tropical biodiversity assets and complementary agricultural biotechnology expertise not available within the EU alone. Academic-to-industry, industry-to-academia, North-to-South, and South-to-North secondments all occur within a single coherent programme, generating a circular knowledge-sharing dynamic that no bilateral or purely European partnership could replicate. This configuration is the direct reason why the exchange of staff is indispensable to achieving the scientific objectives of AGRISWAP.

3.3 Milestones

ID	Title	WP	Due	Verification
MS1	Consortium Agreement and Data Management Plan finalised	WP5	M3	Signed CA submitted to Project Officer; DMP uploaded to grant management system
MS2	First cohort of secondments initiated (at least 5 staff in mobility)	WP1, WP2	M9	Secondment certificates and host-institution confirmation letters
MS3	Pilot-scale fermentation protocol validated for first target ingredient	WP2	M30	Technical report with process data; reviewed by Scientific Coordination Committee
MS4	Mid-term review completed; secondment plan updated if necessary	WP5	M24	Mid-term review report approved by Supervisory Board
MS5	Stakeholder Workshop 2 held; policy brief drafted	WP3, WP4	M36	Workshop proceedings and attendance list; policy brief circulated for external review
MS6	Joint Research Agreement signed by all partners	WP5	M42	Signed JRA submitted to coordinator; copy provided to Project Officer

3.4 Deliverables

ID	Title	WP	Due	Type	Dissemination
D1.1	Standardised Strain-Engineering Protocol Handbook	WP1	M30	Report	Public
D2.1	Pilot-Scale Fermentation Optimisation Report (two target	WP2	M42	Report	Confidential

	ingredients)				
D3.1	Regulatory Pathway Roadmap for Precision-Fermentation Ingredients (EU Novel Food)	WP3	M40	Report	Public
D3.2	Policy Brief: Regulatory Implications of Precision Fermentation for EU Food Safety Agencies	WP3	M42	Other	Public
D4.1	Project Website (launch)	WP4	M3	Websites, patents filing, videos etc.	Public
D4.2	Stakeholder Workshop Reports (x2)	WP4	M38	Report	Public
D5.1	Data Management Plan (initial and updated)	WP5	M6	Data Management Plan	Sensitive
D5.2	Final Project Report including Secondment Statistics and Gender-Action Plan Assessment	WP5	M48	Report	Public

3.5 Resources and Budget Summary

Cost Category	Amount (EUR)
Personnel Costs	0
Subcontracting	0
Travel & Subsistence	0
Equipment	0
Other Direct Costs	756000
Indirect Costs (25%)	324000
TOTAL	1080000

3.6 Risk Management

ID	Risk	Likelihood	Impact	Mitigation
R1	Mobility disruption: visa delays or travel restrictions prevent planned secondments, particularly for Brazil –EU exchanges.	Medium	High	Secondment plan includes buffer months and allows splitting of secondments into shorter stays; alternative host-partner pairings are pre-agreed; visa processes initiated at least 3 months in advance.
R2	IP conflict: disagreement over ownership of jointly	Medium	High	Consortium Agreement with detailed IP provisions

	generated foreground IP delays exploitation and strains partner relations.			(background/foreground und distinction, licensing terms) is concluded before Month 3; FermenTech leads an IP audit at Month 12 and Month 36.
R3	Staff turnover at SME partner: key technical staff at FermenTech Italia leave during the project, reducing industrial expertise available for secondments.	Low	Medium	At least two FermenTech staff per thematic area are trained in each protocol; knowledge is documented in the shared laboratory notebook; coordinator is notified within one month of any key-staff departure.
R4	Scientific underperformance: precision-fermentation targets for one ingredient are not achieved at pilot scale within the planned timeline.	Low	Medium	Scientific Coordination Committee reviews progress quarterly; fallback ingredient options are identified in the secondment plan; TRL targets are revised downward at mid-term review if necessary rather than suppressing outputs entirely.
R5	Gender imbalance: recruitment pipeline fails to achieve the 40% target for the under-represented gender among seconded staff.	Medium	Low	Gender-action plan is activated from Month 1; sending institutions are contractually obliged to report gender composition of secondment nominees; targeted outreach and mentoring are applied where imbalance is detected at the annual review.